
Matthew Samuels

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EDUCATION

Worcester Polytechnic Institute (WPI), Worcester, MA

B.S.+M.S. in Computer Science, Minor in ECE, GPA 3.94 *Expected Graduation December 2026*

Awards: Upsilon Pi Epsilon Honors Society, Dean's list Fall 2022 - Spring 2026

Relevant Coursework: Algorithms, Operating Systems, Database Systems, Software Engineering, Artificial Intelligence, Machine Learning, Programming Language Implementation, Object-Oriented Design, Theoretical Computer Science, Generative AI, Computer Networks, Data Security

Skills

Primary Languages: Java, Python, C, C#, Racket, SQL, HTML, CSS, JavaScript/TypeScript

Auxiliary Languages: Bash, Scala, Rust, R, Haskell

Software/Frameworks: Git, React, Node.js, MongoDB, Express.js, Docker, AWS (Lambda, S3, API Gateway), MATLAB, Microsoft Excel, Figma, Visual Studio, Eclipse, IntelliJ, Unity, KiCad

Projects

Dynamic Programming Code Generation

August 2025 - March 2026

- Developed features for a Scala-based code generation framework that automatically produces implementations of dynamic programming algorithms.
- Designed a model to represent dynamic programming problems and generalize families.
- Built functional tool capable of generating dynamic programming solutions.

Competitive Coding App | github.com/mlsamuels/run-defined

March 2026 - Present

- Developed a full-stack MERN web application that allows users to submit coded computer agents for competitive games and automatically ranks submissions based on match results.
- Engineered the application using MongoDB, Express.js, React, and Node.js, following industry-standard development practices, and published the source code on GitHub.
- Designed and implemented a secure execution environment that safely runs untrusted user-submitted code in isolated Docker containers.
- Deployed and maintained the app on a self-hosted Raspberry Pi server: rundefined.ddns.net

EXPERIENCE

Acend - Platforming Video Game Developer

December 2019 - April 2022

- Developed and launched a fully integrated 10 level platformer using Unity and C#
- Spearheaded all aspects of production, including coding, level architecture, and art design, resulting in a cohesive and polished final product.
- Garnered over 40,000 views on newgrounds.com.

SummerTech - Purchase, NY - Academic Director

June 2022 - July 2026

- Expanded and implemented a programming curriculum covering core computer science concepts, including recursion, algorithms, and data structures, using Java and Python.
- Led a team of 60+ instructors, fostering their technical expertise and teaching effectiveness.
- Managed staff by coordinating teacher assignments, resolving technical challenges, evaluating performance, and supporting day-to-day camp operations.